SEMINAR REPORT

A seminar Report on

IoT in Smart Homes

Submitted in partial fulfillment for the award of

THIRD YEAR DIPLOMA IN

COMPUTER ENGINEERING

By

Sr. No	Enrolment Number	Name of student
1	24610960188	Snehal Manoj Chaudhari
2	24610960190	Ragini Devendrasingh Chauhan
3	24610960233	Hemangi Sujit Patil
4	24610960236	Janhavi Dagaji Patil

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DEPARTMENT OF COMPUTER ENGINEERING

S.S.V.P.S's Bapusaheb Shivajirao Deore Polytechnic, Dhule

(Institute Code : 0059)

S.S.V.P.S's Bapusaheb Shivajirao Deore Polytechnic, Dhule

(An Affiliated Institute of Maharashtra State Board of Technical Education)

DEPARTMET OF COMPUTER ENGINEERING

CERTIFICATE

Sr. No	Enrollment Number	Name of student
1	24610960188	Snehal Manoj Chaudhari
2	24610960190	Ragini Devendrasingh Chauhan
3	24610960233	Hemangi Sujit Patil
4	24610960236	Janhavi Dagaji Patil

of **CO5K** Semester of Diploma in **Computer Engineering** of **S.S.V.P.S's B. S. Deore Polytechnic, Dhule** (Inst.Code: 059) has completed the term work satisfactorily in Course **Seminar Presentation and Project Initiation (Course Code: 315003)** for the academic year **2025 to 2026** as prescribed in the curriculum.

Place : Dhule

Date : .....

Signature:

Name: Miss Shraddha Hajari

Subject Teacher

Signature:

Name: Mr. Nishad Patel

Head of the Department

Seal of Institution

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ABSTRACT

The Internet of Things (IoT) connects home sensors, appliances and controllers so that they can sense, communicate and act automatically. In a smart home, devices such as lights, locks, cameras, thermostats and plugs exchange data over a network and are controlled from a mobile app, a voice assistant or a dashboard. This seminar studies the five-layer architecture of an IoT smart home, the common sensors, actuators and controllers, and the main communication protocols: Wi-Fi, Bluetooth LE, Zigbee, Z-Wave, Thread and the Matter standard. It also reviews applications in lighting and comfort, security, energy saving and health monitoring of the elderly, together with the advantages and disadvantages of smart homes.

The main open problems are security, privacy and interoperability between vendors. Weak default passwords, unpatched firmware and dependence on vendor clouds expose users to attacks, while devices from different brands often do not work together. Matter, regular firmware updates, a separate IoT network and local processing are discussed as ways to reduce these problems. The report ends with a project initiation plan for a local, low-cost smart home controller built with ESP32 sensor and relay nodes, MQTT and a single Raspberry Pi dashboard that works without the cloud.

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CHAPTER 1

INTRODUCTION

1.1 Smart Home and IoT

A smart home is a house in which lights, locks, cameras, thermostats and other appliances are connected to a network so that they can be monitored and controlled automatically or from a phone. The Internet of Things (IoT) makes this possible. An IoT device is a small device with a sensor, a processor and a network link that collects data and exchanges it with other devices and with software.

1.2 How a Smart Home Works

A typical smart home works in four steps:

1. **Sense:** Sensors collect data such as temperature, motion, door status and power use.
2. **Communicate:** A hub or router carries the data over Wi-Fi, Bluetooth, Zigbee, Z-Wave or Thread.
3. **Decide:** Software applies rules to the data, either in the cloud or locally.
4. **Act:** Actuators such as lights, locks and thermostats act on the decision, and the user can also control them from an app.

Figure 1.1 shows this cycle.

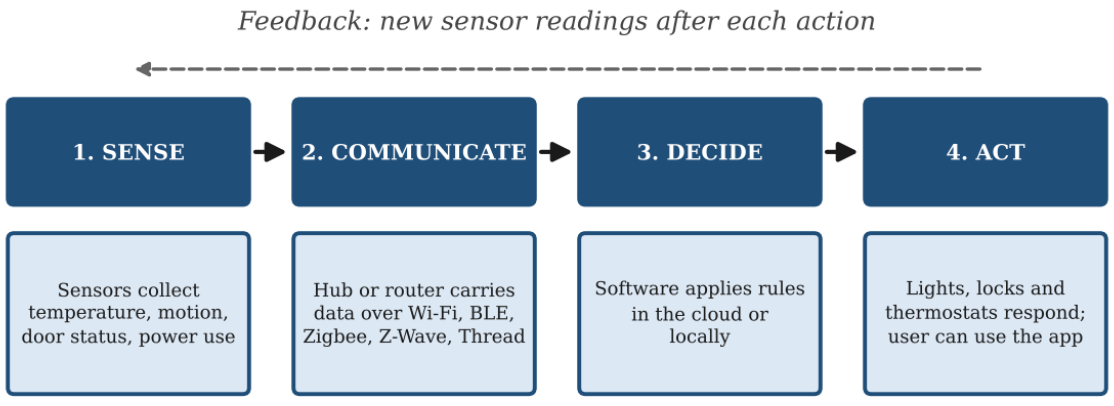


Figure 1.1: Four-step working cycle of a smart home (sense, communicate, decide, act)

1.3 Why It Matters

The home is the place where most people first meet IoT, so the quality of smart home devices affects ordinary users directly. Early devices had weak security, such as default passwords and no firmware updates, and devices from different brands often do not work together. These problems of security, privacy and interoperability are the reason this seminar studies IoT in smart homes.

1.4 Organisation of the Report

This report presents a literature survey of related work, the architecture, devices, protocols and applications of smart homes, their advantages and disadvantages, and a conclusion. It ends with a project initiation plan for a local, low-cost smart home controller.

CHAPTER 2
LITERATURE REVIEW

The following studies were reviewed to understand the basics of IoT in smart homes, the communication technologies used and the problems that remain open. Table 2.1 summarises the main finding of each paper.

1. "Introduction to IoT" is a beginner-level guide that explains IoT from scratch: sensors collect data, a network carries it, and the cloud and apps present it. It builds no system and uses no dataset, and it names security and privacy as the biggest open problems.

2. "An IoT-Based Smart Home Automation System" compares Bluetooth, Zigbee and Wi-Fi for home automation and build a low-cost smart home that is controlled from a phone app. The work is a small home prototype, and security and interoperability between brands are not its focus.

3. "IoT-based Smart Home Monitoring and Automation System Using Android Application" builds a home monitoring and automation system with sensors, a microcontroller, Wi-Fi and an Android app that shows status and alerts. It depends on Wi-Fi and the phone app, with no local dashboard or offline operation.

4. "Design and Implementation of Wi-Fi based Smart Home Automation System" uses a NodeMCU, relays and Wi-Fi to switch home devices cheaply, and compares Bluetooth and Zigbee as alternatives. The control is basic on/off, with limited security and few automation rules.

5. "Literature Survey for IoT-based Smart Home Automation: A Comparative Analysis" presents a comparative survey of 20 IoT smart home works, comparing them by issue addressed, merit, demerit, communication, controller, storage and tools. Most compared systems are Wi-Fi and Arduino based and many list no demerits; being a review, it builds no system.

6. "IoT Based Advanced Home Automation" builds a low-cost home automation system with voice and website control, using a PIC16F877A, GSM SIM900A, Bluetooth HC-05 and an Android voice app, with gas, burglar and water-level safety for elderly and disabled users.

It is semi-automated, Bluetooth works only offline, and only eight fixed voice commands are supported.

7. “Smart Home Automation System Based on IoT” proposes a scalable architecture with an ESP32 NodeMCU, four relays and the Blynk app for remote control, monitoring and rule-based automation of lights and a fan. The work is largely conceptual, with no test results, and the security methods it describes (TLS, OAuth) are not implemented.

8. “IoT-Based Smart Fire and Gas Safety System for Enhanced Household Security in Rural Environments” designs a low-cost fire and gas safety system for rural homes using an ESP32, MQ2 and MQ5 gas sensors, a DHT11 and a buzzer with Blynk alerts. The buzzer sounds in about 2.4 s and the phone alert arrives in about 3.5 s, at 0.5 to 1 W power. It was tested only in a controlled kitchen, the MQ sensors are cross-sensitive and the DHT11 is low precision.

9. “The Impact of IoT on Smart Home Energy Management” studies how IoT monitoring and automation reduce home energy use and cost, using smart meters, thermostats, smart appliances, Python analysis and an Arduino-based Homergy box. Their results show a fall from 500 to 400 kWh per month (20%), but the dataset is assumed, and privacy, security and interoperability remain open.

10. “IoT Based Home Automation System Using ThingSpeak” controls lights, fans, a sprinkler and a door lock remotely and monitor energy use with an Arduino UNO, Raspberry Pi 3, the ThingSpeak cloud and an Android app. Monthly kWh readings from January to May 2019 were lower with the system, but it needs continuous power and internet, has no battery backup and depends on ThingSpeak.

11. “Smart Home Switch System Using IoT” builds a smart home switch system with an ESP32, ACS712 current sensor, DHT11 and an 8-channel relay, controlling four loads from a phone through Blynk while showing current, power, temperature and humidity. Power is estimated by assuming a constant 240 V supply, the DHT11 is low precision, and the system depends on Wi-Fi and the cloud with no quantified results.

12. “Internet of Things (IoT): BLYNK Framework for Smart Home” uses a Raspberry Pi as server and NodeMCU clients with a DHT11 and a relay, controlled by a Blynk phone app,

to retrofit an existing house cheaply with wireless nodes and no rewiring. The prototype is very small (one sensor, one relay), NodeMCU I/O is limited, backup power is needed and security is not studied.

Sr. No.	Approach Type	Technology Used	Dataset	Key Objective	Limitations
1	Introductory IoT guide (Ananna & Saifuzzaman, 2023)	Sensors, network, cloud and apps (general IoT architecture)	None; a survey-style guide, no experimental dataset	To explain IoT from scratch for beginners.	Names security and privacy as the big open problems; no system is built or tested.
2	Low-cost smart home automation (Stolojescu-Crisan et al., 2021)	Bluetooth, Zigbee and Wi-Fi compared; phone app	Sensor and control data from a small home prototype	To compare Bluetooth, Zigbee and Wi-Fi and build a low-cost smart home controlled from a phone app.	Small home prototype; security and interoperability between brands are not the focus.
3	Smart home monitoring with Android app (Akter et al., 2023)	Sensors, microcontroller, Wi-Fi, Android app	Sensor readings from the prototype	To monitor and automate a home, showing status and alerts in a mobile app.	Depends on Wi-Fi and a phone app; no local dashboard or offline operation.
4	Wi-Fi based home automation (Sachdeva, 2018)	NodeMCU, relays, Wi-Fi (Bluetooth and Zigbee compared)	Switching and control data from the prototype	To switch home devices cheaply through relays using NodeMCU and Wi-Fi.	Basic on/off control over Wi-Fi; limited security and automation rules.

Sr. No.	Approach Type	Technology Used	Dataset	Key Objective	Limitations
5	Comparative literature survey of 20 IoT smart home works (Kaur et al., n.d.)	Arduino, Raspberry Pi, ESP8266, ZigBee, Wi-Fi, GSM; web and mobile apps (Blynk)	None; compares earlier studies in two tables	To compare IoT smart home systems by issue addressed, merit, demerit, communication, controller, storage and tools.	Review only; no system is built; many compared works list no demerits; mostly Wi-Fi and Arduino based.
6	Voice and web controlled home automation (Nadgaundi et al., 2020)	PIC16F877A, GSM SIM900A, Bluetooth HC-05, PHP and MySQL web page, SS Voice Android app	None; sensor status (LPG, PIR, IR, water level) logged online	To build a low-cost system with voice and website control plus gas, burglar and water-level safety, helpful for elderly and disabled users.	Semi-automated, needs a user command; Bluetooth works only in offline mode; only eight fixed voice commands.
7	Smart home automation system based on IoT (Kadiyan et al., 2023)	ESP32 NodeMCU, four relays, Blynk app, Wi-Fi	None; proposed architecture and block diagram	To propose a scalable architecture for remote control, monitoring and rule-based automation of lights and a fan.	Largely conceptual; no test results; security (TLS, OAuth) is described but not implemented.
8	IoT smart fire and gas safety system for rural homes (Sharma et al., 2026)	ESP32, MQ2, MQ5, DHT11, buzzer, Blynk	Readings from a controlled kitchen test; no public dataset	To give low-cost real-time smoke and LPG leak alerts; buzzer in about 2.4 s, phone alert in about 3.5 s, 0.5 to 1 W power.	Tested only in a controlled kitchen; MQ sensors are cross-sensitive and need warm-up; DHT11 is low precision; remote alerts need Wi-Fi.

Sr. No.	Approach Type	Technology Used	Dataset	Key Objective	Limitations
9	Impact of IoT on smart home energy management (Kumar & Pandey, 2023)	Smart meters, thermostats, smart appliances; Python analysis; Homergy box (Arduino Mega, NodeMCU, relays)	Assumed CSV of home energy use before and after IoT	To study how IoT monitoring and automation reduce energy use and cost; reports 500 to 400 kWh per month (20%).	Dataset is assumed, so results are illustrative; privacy, security and interoperability remain open.
10	Home automation using ThingSpeak (Akhtar et al., 2019)	Arduino UNO, Raspberry Pi 3, ThingSpeak cloud, Android app, PIR, LDR, soil moisture sensor, servo lock	ThingSpeak logs and monthly kWh readings, January to May 2019, with and without the system	To control lights, fans, sprinkler and door lock remotely and monitor energy use; energy use was lower with the system.	Needs continuous power and internet; no battery backup; depends on the ThingSpeak cloud.
11	Smart Home Switch System using IoT (Ab Rahim & Yahaya, 2023)	ESP32, ACS712 current sensor, DHT11, 8-channel relay, Blynk	None; live current, power, temperature and humidity readings	To control four loads from a phone and monitor current, power and room conditions.	Power is estimated assuming a constant 240 V supply; DHT11 is low precision; cloud and Wi-Fi dependent; no quantified results.
12	Blynk framework for smart home (Media's et al., 2019)	Raspberry Pi as server, NodeMCU clients, DHT11, relay, Blynk	None; prototype with event logs stored on the Raspberry Pi	To retrofit an existing house cheaply with wireless nodes and no rewiring, controlled by a Blynk phone app.	Very small prototype (one sensor, one relay); NodeMCU I/O load is limited; needs backup power; security not studied.

Table 2.1: Summary of literature reviewed

CHAPTER 3

INFORMATION

3.1 Five-Layer Architecture

A smart home IoT system is usually described in five layers:

- **Perception layer:** sensors and actuators.
- **Network layer:** Wi-Fi, Bluetooth, Zigbee, Z-Wave and Thread.
- **Middleware layer:** gateway or cloud, rules and device management.
- **Application layer:** mobile app, voice assistant and dashboard.
- **Business layer:** energy billing and maintenance services.

Figure 3.1 shows the five layers and the main items in each.

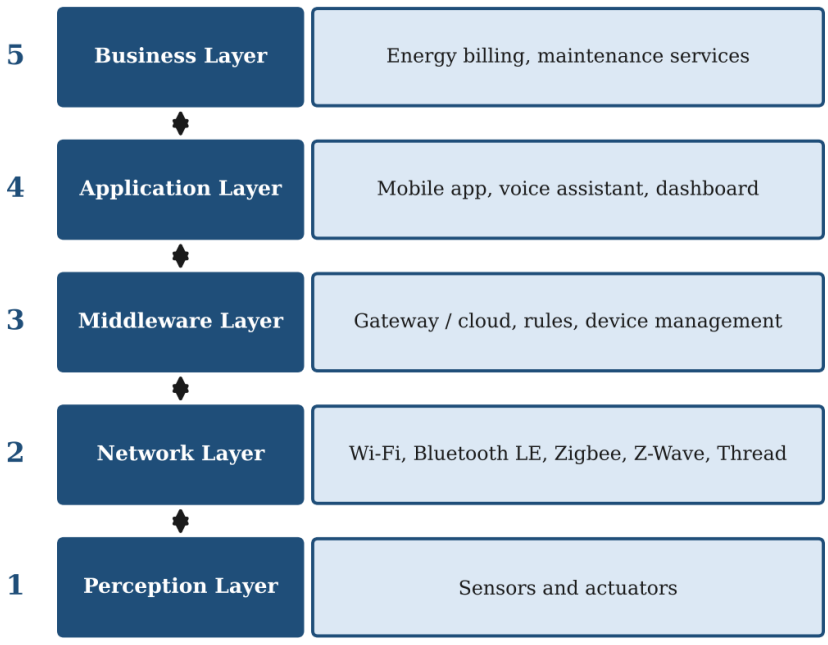


Figure 3.1: Five-layer architecture of an IoT smart home

3.2 Devices

- **Sensors:** temperature, motion, door, smoke and water leak.
- **Actuators:** bulbs, plugs, locks and thermostats.
- **Controllers:** hub, smart speaker and Raspberry Pi.

Figure 3.2 shows how these devices connect through a hub in a typical smart home.

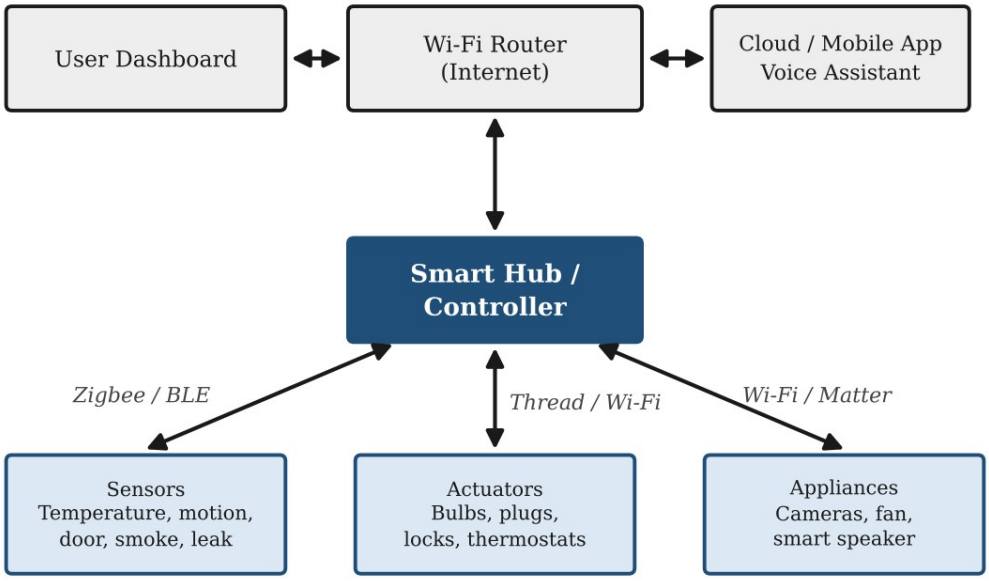


Figure 3.2: Typical smart home network with hub, devices and cloud

3.3 Protocols and the Matter Standard

Devices in a smart home talk to each other using several protocols, each suited to a different need. Matter is a newer standard that runs over Wi-Fi and Thread so that devices from different brands can work together. Table 3.1 compares them.

Protocol	Type	Notes
Wi-Fi	High bandwidth IP	Cameras, plugs; higher power use
Bluetooth LE	Short range	Device setup, wearables
Zigbee	Low-power mesh, IEEE 802.15.4	Common in lighting
Z-Wave	Low-power mesh	Strong security design
Thread	IPv6 low-power mesh	No single point of failure
Matter	Standard over Wi-Fi and Thread	Cross-brand interoperability; 1.4 added energy devices

Table 3.1: Smart home communication protocols

3.4 Applications

- **Lighting and comfort:** schedules, motion lights and voice control.
- **Security:** locks, cameras, door sensors and phone alerts.
- **Energy:** smart plugs and thermostats; 10 to 23 percent saving has been reported for learning thermostats.
- **Health:** fall detection and activity monitoring for the elderly.

3.5 Security and Privacy Risks

- Buffer overflows (28.25%) and DoS flaws (27.20%) are the most common vulnerabilities.
- Default passwords still enable botnets such as Mirai.
- 76% of users know the risks, but only 24% update firmware regularly.

Figure 3.3 plots the vulnerability and user-awareness figures quoted above.

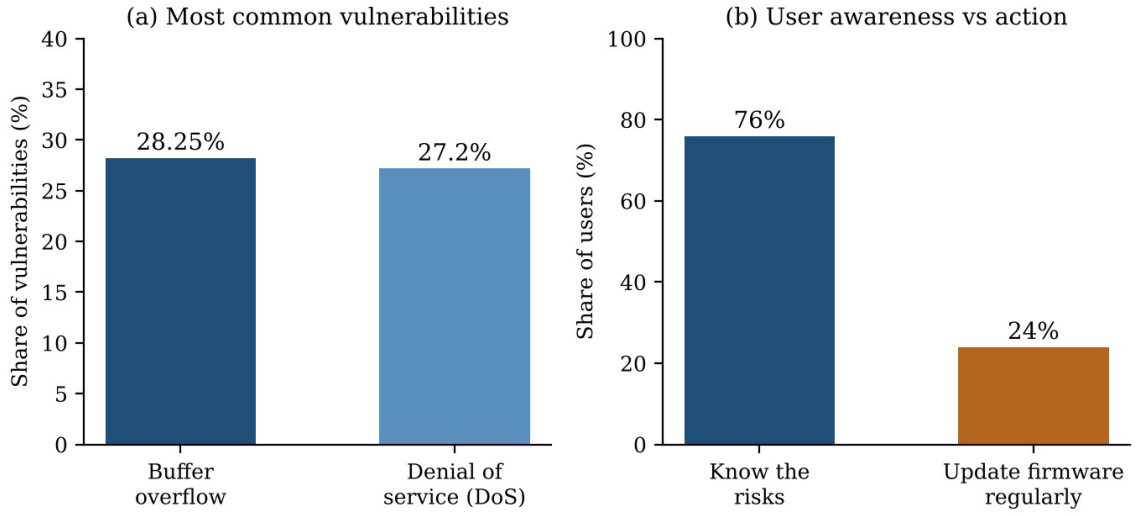


Figure 3.3: Common smart home vulnerabilities and user behaviour

Defence: change default passwords, update firmware regularly, keep IoT devices on a separate network and use local processing where possible.

CHAPTER 4

ADVANTAGES AND DISADVANTAGES

Smart homes bring clear benefits to daily life, but they also create new risks and costs. Table 4.1 lists the main advantages and disadvantages found in this study.

Advantages	Disadvantages
Convenience: one app or voice control	Security risks: default passwords, unpatched firmware
Energy saving through schedules and thermostats	Privacy: usage data reveals when people are home
Safety: smoke, gas and leak alerts	Interoperability gaps between brands
Helps elderly and disabled people	Device and hub cost; vendor support ends
Automation without effort	Depends on internet and power; cloud outages

Table 4.1: Advantages and disadvantages of IoT in smart homes

Most of the disadvantages, namely security, privacy and interoperability, can be reduced by using the Matter standard, regular firmware updates, a separate IoT network and local processing, as discussed in Chapter 3.

CHAPTER 5
CONCLUSION

IoT makes homes more convenient, safer and more energy efficient. Sensors, actuators and controllers working together let a home sense, decide and act automatically, with control from a phone app, a voice assistant or a dashboard.

The studies reviewed agree that security, privacy and interoperability are the main weaknesses of smart homes. Default passwords, unpatched firmware and gaps between brands put users at risk. The Matter standard, regular firmware updates, a separate IoT network and local processing help to address these problems.

A simple, local, low-cost system is a practical next step. The project initiation plan in Chapter 6 proposes such a system, built with ESP32 nodes, MQTT and a single dashboard that works without the cloud.

CHAPTER 6

PROJECT INITIATION

The proposed project is a local, low-cost smart home controller. ESP32 sensor and relay nodes report over Wi-Fi using MQTT to one dashboard on a Raspberry Pi, and the system works without the cloud.

Figure 6.1 shows the block diagram of the proposed system.

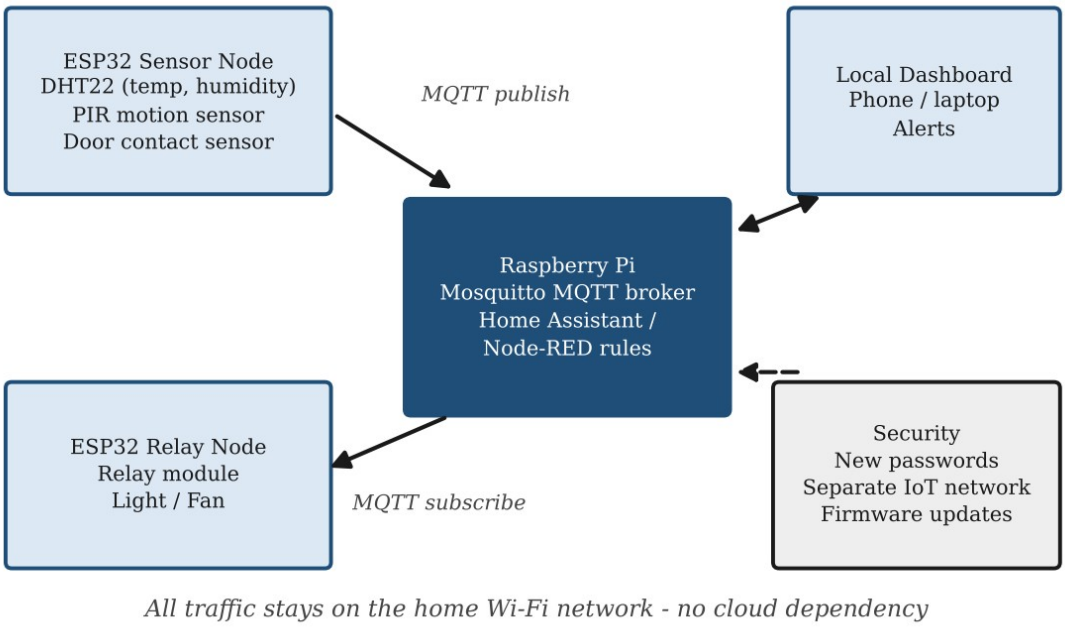


Figure 6.1: Block diagram of the proposed local smart home controller

6.1 Description of Problem Statement

Most commercial smart home devices depend on vendor clouds. Their default settings are weak and devices from different brands do not work together. Home users and students lack a simple, low-cost, private system that is easy to extend.

Problem statement: build a local smart home with sensor and relay nodes and one dashboard, without cloud dependency.

6.2 Scope and Objectives

Scope: one room or flat, with sensing (temperature, humidity, motion, door), control (lights, fan), and a dashboard with alerts. Cameras, voice assistants and cloud services are out of scope.

Objectives:

- Study IoT architecture, protocols and security risks.
- Build sensor and relay nodes reporting to a local controller.
- Provide a dashboard and simple automation rules.
- Apply basic security: new passwords, separate network and updates.
- Test reliability and response time, and document the results.

6.3 Stakeholders

Stakeholder	Role	Interest
Student	Designer and builder	Complete seminar and project
Guide (faculty)	Supervisor	Technical quality, timely progress
Department / HOD	Approver	Course outcomes of SPI - 315003
Home users (family)	End users	Easy control, safety, privacy

Table 6.1: Stakeholders

6.4 Seminar Execution Plan

Week	Activity
1	Topic selection and literature survey
2	Gap, problem statement and objectives
3	Prepare report and presentation; guide review
4	Deliver seminar and collect feedback

Table 6.2: Seminar execution plan

6.5 Platform / Equipment / Resources Identification

Type	Items
Hardware	ESP32 boards, DHT22 temperature sensor, PIR motion sensor, door contact sensor, relay module
Controller	Raspberry Pi or laptop on home Wi-Fi
Software	Mosquitto MQTT broker, Home Assistant or Node-RED, Arduino IDE or MicroPython
Tools	Breadboard, jumper wires, multimeter, phone for the dashboard

Table 6.3: Platform, equipment and resources

CHAPTER 7

SEMINAR PRESENTATION

	S.S.V.P.S's B. S. Deore Polytechnic, Dhule **Department of Computer Engineering** *Academic Year : 2026-2027*

SEMINAR AND PROJECT INITIATION COURSE

Sem : CO 5 K	**SPI – 315003**	**Date :**

Topic : IoT in Smart Homes

Roll No.	Name of Student
20	Snehal Manoj Chaudhari
21	Ragini Devendrasingh Chauhan
54	Hemangi Sujit Patil
57	Janhavi Dagaji Patil

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5. Conclusion
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   - 6. 2 Scope and objectives.
   - 6. 3 Stake holder
   - 6. 4 Seminar execution plan
   - 6.5 Platform/ Equipment/ Resources identification.
7. Bibliography
8. References

Abstract

- IoT connects home sensors, appliances and controllers so they can sense, communicate and act automatically
- Seminar covers architecture, devices, protocols and applications
- Main open problems: security, privacy and interoperability between vendors
- Matter standard unifies devices over Wi-Fi and Thread
- Ends with a project initiation plan for a local, low-cost smart home
- **Keywords: IoT, smart home, Matter, Zigbee, security**

1. Introduction

- Smart home: lights, locks, cameras, thermostats and appliances connected to a network
- IoT: small devices with sensors, a processor and a network link that exchange data
- **How it works**
  - Sensors collect data (temperature, motion, door, power)
  - Hub or router carries it over Wi-Fi, Bluetooth, Zigbee, Z-Wave or Thread
  - Software applies rules in the cloud or locally
  - Actuators act: lights, locks, thermostat; user controls it from an app
- Why it matters: homes are where most people meet IoT, early devices had weak security, brands often do not interoperate

2. Literature Survey (1/4)

Sr. No.	Approach Type	Technology Used	Dataset	Key Objective	Limitations
1	Introductory IoT guide (Ananna & Saifuzzaman, 2023)	Sensors, network, cloud and apps (general IoT architecture)	None; a survey-style guide, no experimental dataset	To explain IoT from scratch for beginners.	Names security and privacy as the big open problems; no system is built or tested.
2	Low-cost smart home automation (Stolojescu-Crisan et al., 2021)	Bluetooth, Zigbee and Wi-Fi compared; phone app	Sensor and control data from a small home prototype	To compare Bluetooth, Zigbee and Wi-Fi and build a low-cost smart home controlled from a phone app.	Small home prototype; security and interoperability between brands are not the focus.
3	Smart home monitoring with Android app (Akter et al., 2023)	Sensors, microcontroller, Wi-Fi, Android app	Sensor readings from the prototype	To monitor and automate a home, showing status and alerts in a mobile app.	Depends on Wi-Fi and a phone app; no local dashboard or offline operation.

2. Literature Survey (2/4)

Sr. No.	Approach Type	Technology Used	Dataset	Key Objective	Limitations
4	Wi-Fi based home automation (Sachdeva, 2018)	NodeMCU, relays, Wi-Fi (Bluetooth and Zigbee compared)	Switching and control data from the prototype	To switch home devices cheaply through relays using NodeMCU and Wi-Fi.	Basic on/off control over Wi-Fi; limited security and automation rules.
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2. Literature Survey (3/4)

Sr. No.	Approach Type	Technology Used	Dataset	Key Objective	Limitations
7	Smart home automation system based on IoT (Kadiyan et al., 2023)	ESP32 NodeMCU, four relays, Blynk app, Wi-Fi	None; proposed architecture and block diagram	To propose a scalable architecture for remote control, monitoring and rule-based automation of lights and a fan.	Largely conceptual; no test results; security (TLS, OAuth) is described but not implemented.
8	IoT smart fire and gas safety system for rural homes (Sharma et al., 2026)	ESP32, MQ2, MQ5, DHT11, buzzer, Blynk	Readings from a controlled kitchen test; no public dataset	To give low-cost real-time smoke and LPG leak alerts; buzzer in about 2.4 s, phone alert in about 3.5 s, 0.5 to 1 W power.	Tested only in a controlled kitchen; MQ sensors are cross-sensitive and need warm-up; DHT11 is low precision; remote alerts need Wi-Fi.
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2. Literature Survey (4/4)

Sr. No.	Approach Type	Technology Used	Dataset	Key Objective	Limitations
10	Home automation using ThingSpeak (Akhtar et al., 2019)	Arduino UNO, Raspberry Pi 3, ThingSpeak cloud, Android app, PIR, LDR, soil moisture sensor, servo lock	ThingSpeak logs and monthly kWh readings, January to May 2019, with and without the system	To control lights, fans, sprinkler and door lock remotely and monitor energy use; energy use was lower with the system.	Needs continuous power and internet; no battery backup; depends on the ThingSpeak cloud.
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12	Blynk framework for smart home (Media's et al., 2019)	Raspberry Pi as server, NodeMCU clients, DHT11, relay, Blynk	None; prototype with event logs stored on the Raspberry Pi	To retrofit an existing house cheaply with wireless nodes and no rewiring, controlled by a Blynk phone app.	Very small prototype (one sensor, one relay); NodeMCU I/O load is limited; needs backup power; security not studied.

3. Information: architecture, devices

- **Five-layer architecture**
  - Perception: sensors and actuators
  - Network: Wi-Fi, Bluetooth, Zigbee, Z-Wave, Thread
  - Middleware: gateway or cloud, rules, device management
  - Application: mobile app, voice assistant, dashboard
  - Business: energy billing, maintenance services
- **Devices**
  - Sensors: temperature, motion, door, smoke, water leak
  - Actuators: bulbs, plugs, locks, thermostats; Controllers: hub, speaker, Raspberry Pi

3. Information: protocols, Matter

Protocol	Type	Notes
Wi-Fi	High bandwidth IP	Cameras, plugs; higher power use
Bluetooth LE	Short range	Device setup, wearables
Zigbee	Low-power mesh, IEEE 802.15.4	Common in lighting
Z-Wave	Low-power mesh	Strong security design
Thread	IPv6 low-power mesh	No single point of failure
Matter	Standard over Wi-Fi and Thread	Cross-brand interoperability; 1.4 added energy devices

3. Information: applications, security

- **Applications**
  - Lighting and comfort: schedules, motion lights, voice control
  - Security: locks, cameras, door sensors, phone alerts
  - Energy: smart plugs, thermostats (10 to 23 percent saving reported for learning thermostats)
  - Health: fall detection and activity monitoring for the elderly
- **Security and privacy risks**
  - Buffer overflows (28.25%) and DoS flaws (27.20%) are the most common vulnerabilities
  - Default passwords still enable botnets such as Mirai
  - 76% of users know the risks; only 24% update firmware regularly
  - Defence: change passwords, update firmware, separate IoT network, local processing

4. Advantages / Disadvantages

Advantages	Disadvantages
Convenience: one app or voice control	Security risks: default passwords, unpatched firmware
Energy saving through schedules and thermostats	Privacy: usage data reveals when people are home
Safety: smoke, gas and leak alerts	Interoperability gaps between brands
Helps elderly and disabled people	Device and hub cost; vendor support ends
Automation without effort	Depends on internet and power; cloud outages

5. Conclusion

- IoT makes homes more convenient, safer and more energy efficient
- Surveys agree: security, privacy and interoperability are the main weaknesses
- Matter, regular updates and local processing address them
- A simple local low-cost system is a practical next step (Section 6)

6. Project Initiation

- **Proposed project: local, low-cost smart home controller**
  - ESP32 sensor and relay nodes report over Wi-Fi using MQTT
  - One dashboard on a Raspberry Pi; works without the cloud
  - Covers: problem statement, scope and objectives, stakeholders, execution plan, resources

6.1 Description of problem statement

- Most commercial smart home devices depend on vendor clouds
- Default settings are weak and different brands do not work together
- Home users and students lack a simple, low-cost, private system that is easy to extend
- **Problem: build a local smart home with sensor and relay nodes and one dashboard, without cloud dependency**

6.2 Scope and objectives

- **Scope**
  - One room or flat; sensing (temperature, humidity, motion, door); control (lights, fan); dashboard and alerts
  - Out of scope: cameras, voice assistants, cloud services
- **Objectives**
  - Study IoT architecture, protocols and security risks
  - Build sensor and relay nodes reporting to a local controller
  - Provide a dashboard and simple automation rules
  - Apply basic security: new passwords, separate network, updates
  - Test reliability and response time; document results

6.3 Stakeholders

Stakeholder	Role	Interest
Student	Designer and builder	Complete seminar and project
Guide (faculty)	Supervisor	Technical quality, timely progress
Department / HOD	Approver	Course outcomes of SPI - 315003
Home users (family)	End users	Easy control, safety, privacy

6.4 Seminar execution plan

Week	Activity
1	Topic selection and literature survey
2	Gap, problem statement and objectives
3	Prepare report and presentation; guide review
4	Deliver seminar and collect feedback
5 to 6	Component list, circuit design, software setup

6.5 Platform/ Equipment/ Resources

Type	Items
Hardware	ESP32 boards, DHT22 temperature sensor, PIR motion sensor, door contact sensor, relay module
Controller	Raspberry Pi or laptop on home Wi-Fi
Software	Mosquitto MQTT broker, Home Assistant or Node-RED, Arduino IDE or MicroPython
Tools	Breadboard, jumper wires, multimeter, phone for the dashboard

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SEMINAR and PROJECT INITIATION Assessment / Evaluation

Sr. No	Enrollment Number	Seat Number	Name of student
1	24610960188	388507	Snehal Manoj Chaudhari
2	24610960190	388508	Ragini Devendrasingh Chauhan
3	24610960233	388541	Hemangi Sujit Patil
4	24610960236	388544	Janhavi Dagaji Patil

Formative Assessment

Sr. No.	Criteria	Weightage	Exam Seat Number			
			388507	388508	388541	388544
1	Selection Topic / Theme of seminar	5				
2	Literature review and data presentation	5				
3	Quality of Preparation and innovativeness	5				
4	Q-A handling	5				
5	Time Management	5				
6	Seminar Presentation report	10				
7	Selection of Theme of Problem Statement and its innovativeness	5				
8	Stages of development of Action plan	5				
9	Prototyping	5				
	Total	**50**				
	Scaled to 25					

Sign :

Name : Miss Shraddha Hajari

(Course Expert’s)